A Concept Level Energy-Based Framework for Interpreting Black-Box Large Language Model Responses

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Abstract

The widespread adoption of proprietary Large Language Models (LLMs) accessed strictly through closed APIs has created a critical challenge for responsible deployment: a fundamental lack of interpretability. To address this, we propose a model-agnostic, post-hoc attribution interpreter operating at the sentence level. Our approach trains an Energy-Based Model (EBM) as a surrogate to capture the LLM's internal conceptual consistency between prompts and responses. This energy landscape guides the training of a lightweight interpreter network. Uniquely, our interpreter operates as a standalone tool; once trained, it quantifies the influence of prompt sentences on a user-specified target output without requiring further API queries to the LLM. By globally training a local interpreter across diverse inputs, our framework captures broader generation patterns and mitigates instance-specific biases. Experiments demonstrate that our EBM accurately simulates the target LLM, allowing the interpreter to effectively identify the prompt sentences most influential in generating specific target outputs.

1 Introduction

Large Language Models (LLMs) have demonstrated extraordinary performance across complex tasks. Consequently, researchers and developers are rapidly adopting them for diverse applications. However, the critical challenge facing this adoption is a fundamental lack of interpretability. Most powerful LLMs are proprietary and accessed strictly through closed-access APIs. Even when architectures and pre-training datasets are available, their complexity obscures exactly how outputs are generated. In high-stakes domains like medicine and law, this opacity is unacceptable, as experts cannot verify the generated output against domain knowledge or detect hidden biases. This prevents meeting the application-grounded standards for responsible deployment (Doshi-Velez and Kim, 2017).

Post-hoc attribution is a primary approach to addressing this opacity. These methods explain model behavior by identifying an importance vector for input features. Essentially, they measure how much each input feature influences the output within a local neighborhood. However, standard attribution techniques, including white-box and model-agnostic, struggle in the context of LLMs. White-box methods, which rely on gradients or activations, are incompatible with closed APIs. Furthermore, the faithfulness of popular proxies like attention weights has been challenged (Jain and Wallace, 2019). Model-agnostic methods exist (Ribeiro et al., 2016; Lundberg and Lee, 2017; Seyyedsalehi et al., 2022), but typically target discriminative models with well-defined outputs. Interpreting generative models is significantly harder as the problem is fundamentally ill-posed. These models utilize complex representations to produce high-dimensional outputs like text. Therefore, effective explanation is hindered by the output's interactivity and sheer volume (Schneider, 2024).

Alternatives like prompt-based self-explanation (Wei et al., 2022) are similarly problematic; they rely on the same process we seek to verify, leading to circular logic and motivated reasoning. Consequently, models often produce plausible-sounding yet unfaithful confabulations (Turpin et al., 2023). While automated prompt engineering can steer model behavior to mitigate biases, it is unsuitable for interpretation. These methods optimize instructions for pre-defined targets (Zhou et al., 2023; Clemmer et al., 2024), rendering them unusable for ambiguous interpretation tasks.

To address these limitations, we propose a model-agnostic, post-hoc attribution interpreter. We diverge from standard approaches by shifting the resolution from noisy tokens to coherent "concepts." We define a concept as a sentence, which is the smallest unit of language that expresses a complete thought. Our goal is to relate elements of

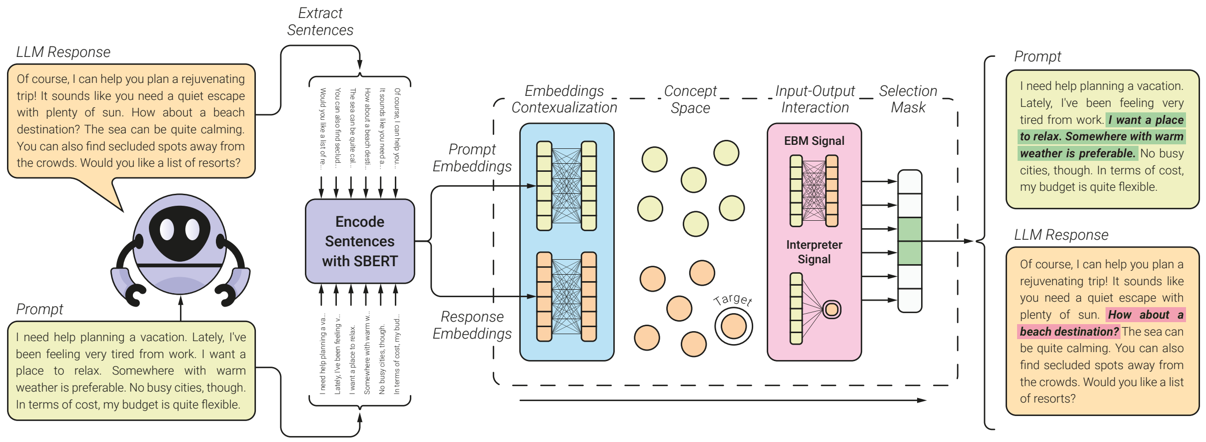

Figure 1: Overview of the proposed framework. The prompt and response of a black-box LLM are split into sentences and embedded via a pre-trained model. An encoding module maps these embeddings to a concept space where proximity reflects relevance. Subsequently, an interaction module evaluates the consistency between input and output concepts to identify the most influential prompt sentences. These modules are trained using signals from an energy network that simulates the generation process of the target LLM.

the output directly to the user prompt at this concept level. Formally, given a prompt $\mathbf{x}$ and an LLM response $\mathbf{y}$, we target a specific subset of the output $\mathbf{y}_T \subset \mathbf{y}$. We then produce an importance vector to identify the subset of prompt sentences $\mathbf{x}_S \subset \mathbf{x}$ that were most influential in generating $\mathbf{y}_T$.

We employ a unique paradigm to globally train a local interpreter. Unlike local interpreters (e.g. LIME (Ribeiro et al., 2016)) which observe only immediate neighborhoods—often causing interpretability illusions (Friedman et al., 2024)—we train across a wider distribution. This enables our model to capture global generation patterns and mitigate intrinsic biases.

Figure 1 illustrates an overview of the approach. We first train a transformer-based Energy-Based Model (EBM) to act as a surrogate for the black-box LLM. It maps the prompt and LLM response sentences to a latent "concept space", which simulates the concept-level relationships embedded in the target LLM, and calculates an energy score. This score quantifies the conceptual consistency between the input and output. We then use this energy landscape to guide the training of the interpreter network. Given a prompt and a target output subset, the interpreter produces an importance vector that isolates the prompt sentences most influential in generating that response. In summary, we:

1. Shift the unit of analysis from tokens to sentences. This enables attribution at the level of semantically coherent concepts, making explanations more intelligible to humans.
2. Introduce a transformer-based EBM capable of learning a random field over prompts and responses. This model effectively distinguishes authentic target-LLM responses from human or other-model text, serving as a robust surrogate for the black-box model.
3. Propose a post-hoc, model-agnostic framework for interpreting black-box LLMs at a conceptual level. This interpreter finds the specific prompt sentences responsible for triggering a target subset of the LLM's output.

2 Related Work

2.1 Post-hoc Attribution Methods

Attribution methods score the importance of input features for a specific model output. White-box approaches to attribution often utilize gradients, propagating output salient signals back to input tokens (Simonyan et al., 2014; Sundararajan et al., 2017; Shrikumar et al., 2017; Chefer et al., 2021). Others use internal attention weights as proxies for feature importance (Xu et al., 2015; Li et al., 2017; Xie et al., 2017; Hao et al., 2021). However, gradients are inaccessible for proprietary APIs, and attention weights are frequently unfaithful to the reasoning process (Jain and Wallace, 2019). Similarly, influence functions pose data-centric explanations (Koh and Liang, 2017) but remain infeasible without access to the base data or model's Hessian.

Perturbation-based methods offer a model-agnostic alternative; they measure output changes when input segments are removed or altered (Ribeiro et al., 2016; Lundberg and Lee, 2017; Yin and Neubig, 2022). Hackmann et al. (2024) apply this to identify influential words in LLM prompts. However, this approach scales poorly for generative tasks. Validating a single explanation often requires thousands of model queries, incurring high computational costs (Enouen et al., 2024; Zhao and Shan, 2024).

Finally, prompt-based self-explanation leverages the LLM's own generation capabilities. Most notably, techniques like Chain-of-Thought (CoT) ask the model to produce a rationale to justify its output (Wei et al., 2022). While compelling, these explanations lack guarantees of faithfulness; they often represent plausible post-hoc rationalizations rather than the true internal computation path (Turpin et al., 2023).

2.2 Energy-Based Models in NLP

Energy-Based Models (EBMs) have been successfully adapted for generative Natural Language Processing (NLP), primarily through learning global scoring functions. Bakhtin et al. (2019) demonstrated that Transformer-based discriminators can function as EBMs; by distinguishing between human and machine text, these models assign low energy to coherent sequences. This established the potential of EBMs as holistic text evaluators.

Additionally, several paradigms utilize EBMs to refine existing model outputs. The Residual EBM approach adds a corrective energy term to the log-probabilities of a base autoregressive model (Deng et al., 2020; Bakhtin et al., 2021). This allows the system to capture high-level properties, such as coherence, that the base model may miss. Alternatively, EBMs act as post-processing rerankers to select the highest-quality result from a set of candidates (Bhattacharyya et al., 2021). Finally, Tu et al. (2020) use a powerful autoregressive teacher to define an energy landscape; a student network is then trained via knowledge distillation to generate outputs that minimize this energy.

2.3 Concept-Based Explanations

Interpretability research is increasingly shifting away from granular token-level attributions and toward concept-based explanations. These methods map decisions to human-intelligible ideas rather than individual features (Kim et al., 2018). Our work aligns with this paradigm by defining a "sentence" as the fundamental conceptual unit, as it represents a robust thought for interpretation.

Treating sentences as semantic objects is well-justified by the history of language modeling. Foundational architectures like BERT used Next Sentence Prediction (NSP) to learn logical relationships (Devlin et al., 2019). Subsequent work on Sentence-BERT confirmed that fine-tuned sentence representations map similar meanings to distinct, nearby points in a vector space (Reimers and Gurevych, 2019). By leveraging sentences as concepts, our work parallels recent architectural innovations such as Large Concept Models (LCMs), which shift the core computational unit from tokens to sentence-level representations (Barrault et al., 2024).

3 Methodology

Our goal is to develop a post-hoc, model-agnostic method for interpreting black-box LLMs, specifically by identifying which input sentences drive the response.. We depart from standard token-level attribution by establishing the sentence as the fundamental unit of analysis. As the smallest linguistic unit expressing a complete proposition, the sentence serves as a robust "concept," enabling us to interpret generation as an interplay of complete ideas rather than ambiguous tokens. Formally, let $\mathbf{x}$ be the prompt and $\mathbf{y}$ be the LLM response. We target a subset of output sentences, $\mathbf{y}_T \subseteq \mathbf{y}$, and seek to quantify the influence of each concept in $\mathbf{x}$ on the generation of $\mathbf{y}_T$.

We propose a two-stage framework to achieve this. First, we pre-train an Energy-Based Model (EBM), $\mathcal{E}_{\text{LM}}(\mathbf{x}, \mathbf{y}; \theta)$, to serve as a differentiable surrogate for the black-box LLM. As a function of θ, this model assigns scalar energy values representing the consistency of a prompt-response pair $(\mathbf{x}, \mathbf{y})$ with the target LLM's generation patterns. Second, we leverage this energy landscape to guide the training of a lightweight interpreter, $\mathcal{IN}(\mathbf{x}, \mathbf{y}_T; \alpha)$, parameterized by α. Taking the prompt, response, and a user-specified target output $\mathbf{y}_T$ as inputs, the interpreter generates a binary vector matching the number of prompt sentences. In this vector, values of 1 identify the subset of prompt sentences $\mathbf{x}_S \subseteq \mathbf{x}$ strictly necessary for generating the target.

3.1 Sentence Extraction and Embedding

The pipeline begins by transforming the input text $\mathbf{x}$ and output text $\mathbf{y}$ into sequences of concepts. We first perform sentence segmentation

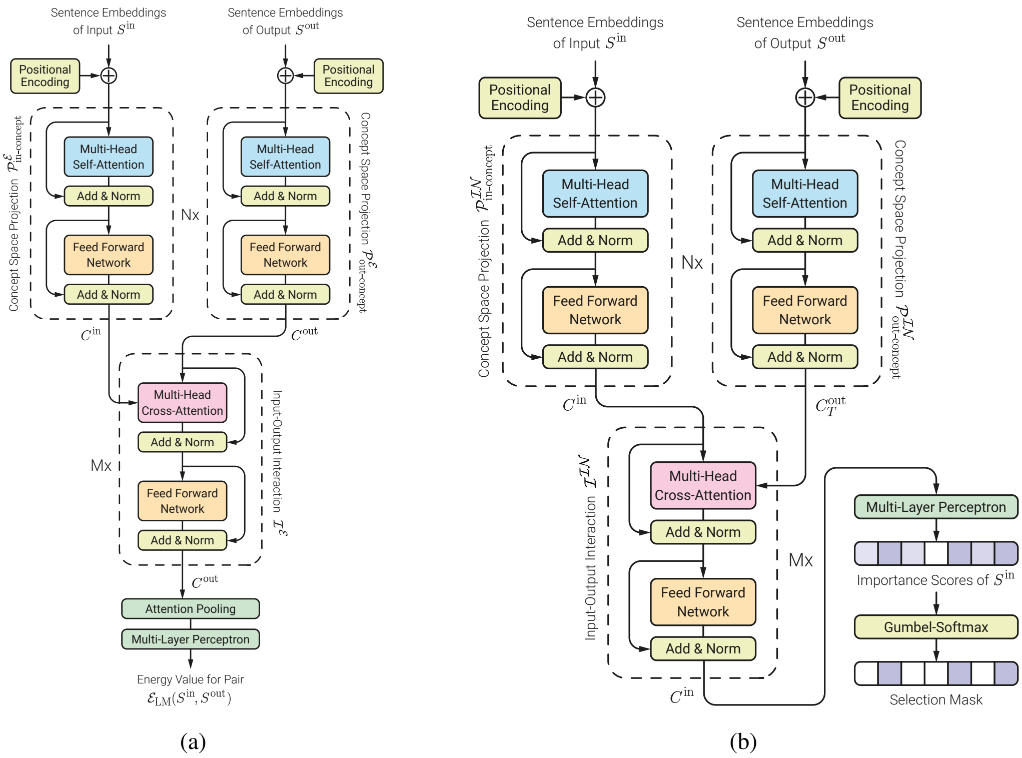

Figure 2: Proposed architectures of (**a**) the energy network and (**b**) the interpreter network

using the spaCy library (Honnibal and Montani, 2017). Subsequently, we employ a frozen, pre-trained Sentence-BERT (Reimers and Gurevych, 2019) module to map each sentence to a fixed-dimensional vector. This yields embedding sequences S^{in} and S^{out}, which function analogously to token embeddings within our architecture. For further preprocessing and implementation details, including padding strategies, visit Appendix A.

3.2 The Energy-Based Surrogate Model

To approximate the black-box LLM's behavior, we design a globally-aware EBM that learns to distinguish variation of authentic prompt-response pairs from corrupted ones; the lower the assigned energy, the more likely the pair is consistent with concept interactions within the LLM. As shown in Figure 2a, the architecture processes sentence embeddings in three stages:

1. **Concept Space Projection:** Static embeddings ($S^{\text{in}}, S^{\text{out}}$) capture meaning in isolation, but lack the specific context of the prompt and response. To remedy this, we pass these embeddings through separate, trainable self-attention modules ($\mathcal{P}^{\mathcal{E}}_{\text{in}}, \mathcal{P}^{\mathcal{E}}_{\text{out}}$) to project them into a dynamic "concept space" ($C^{\text{in}}, C^{\text{out}}$). Here, distances reflect the LLM's internal dependency structure rather than generic semantic similarity. This structure is a function of the model's underlying architecture and training dataset.

2. **Input-Output Interaction:** A cross-attention block allows input concepts C^{in} to attend to output concepts C^{out}. This allows the model to weigh the causal influence of the prompt on the response.

3. **Energy Calculation:** The interacting representations are aggregated via attention pooling and passed through a Multi-Layer Perceptron (MLP) to output a scalar energy $\mathcal{E}_{\text{LM}}(\mathbf{x}, \mathbf{y}; \theta)$.

The EBM is trained in two phases: pre-training via an expanded InfoNCE loss and fine-tuning alongside the interpreter. For pre-training, we generate a dataset of prompt-output pairs $(\mathbf{x}, \mathbf{y})$ from the black-box LLM.

To apply InfoNCE, we generate objective-specific negative samples. For **Fidelity** ($\mathcal{L}_{fidelity}$), negatives are drawn from other LMs and human responses to distinguish the target LLM's style. For **Semantic Choice** ($\mathcal{L}_{choice}$), we use batch-wise off-topic samples: strategy (x_{part}, y'_{part}) distinguishes correct partial responses, while (x'_{part}, y_{part}) identifies correct partial prompts. The model minimizes an adaptive dual-objective loss:

$$\mathcal{L}_{total} = (1-\lambda)\mathcal{L}_{fidelity} + \lambda(\mathcal{L}_{\text{resp\_choice}} + \mathcal{L}_{\text{prompt\_choice}}) \quad (1)$$

where λ is a configurable weight. We define the energy scoring term as $h(\mathbf{u}, \mathbf{v}) = \exp(-\mathcal{E}_{\text{LM}}(\mathbf{u}, \mathbf{v}; \theta)/\tau)$. The InfoNCE losses are formulated as:

$$\mathcal{L} = -\log\left(\frac{h(\mathbf{x}_i, \mathbf{y}_i)}{h(\mathbf{x}_i, \mathbf{y}_i) + \sum_{(\mathbf{x}', \mathbf{y}') \in \mathcal{N}_i} h(\mathbf{x}', \mathbf{y}')}\right) \quad (2)$$

Here, $\mathcal{N}_i$ is the specific negative set (other-LM responses for fidelity; batch negatives for choice). Appendix C details this procedure, which sharpens the discrimination boundary to better guide the interpreter.

3.3 The Interpreter Model

Given prompt $\mathbf{x}$, response $\mathbf{y}$, and target $\mathbf{y}_T \subseteq \mathbf{y}$, the interpreter identifies prompt sentences influential to $\mathbf{y}_T$. It outputs a binary vector where 1 indicates a necessary precursor sentence.

Figure 2b illustrates the architecture. Embeddings S^{in} and S^{out} are projected to concept spaces via self-attention modules. Target concepts C_T^{out} are retained; input concepts C^{in} then attend to them via cross-attention. An MLP and Gumbel-Softmax unit (Jang et al., 2016) (see Appendix B) process the weights to yield a binary importance vector.

Let $\tilde{\mathbf{x}} = \mathbf{x} \odot \mathcal{IN}(\mathbf{x}; \mathbf{y}_T, \alpha)$ be the selected subset. A successful selection minimizes the energy of the authentic pair $(\tilde{\mathbf{x}}, \mathbf{y}_T)$ and maximizes the energy of the irrelevant remainder $(\mathbf{x} - \tilde{\mathbf{x}}, \mathbf{y}_T)$. The interpreter optimization is thus:

$$\underset{\alpha}{\operatorname{argmax}} \;\; \mathbb{E}_{(x,y)}\Big[\mathcal{E}_{\text{LM}}(\mathbf{x} - \tilde{\mathbf{x}},\; \mathbf{y}_T; \theta) - \mathcal{E}_{\text{LM}}(\tilde{\mathbf{x}},\; \mathbf{y}_T; \theta)\Big] \quad (3)$$

3.3.1 EBM-Guided Training of the Interpreter

Masking inputs introduces distribution shifts (Hsia et al., 2023). To mitigate this, we fine-tune the EBM during interpreter training (Figure 6) using an alternating procedure:

1. **Update Interpreter:** For fixed EBM parameters $\theta^{(k-1)}$, we update α to maximize Eq. 3.

2. **Query LLM:** Obtain the response for the masked prompt:

$$\tilde{\mathbf{x}} = \mathbf{x} \odot \mathcal{IN}(\mathbf{x}, \mathbf{y}_T; \alpha^{(k)}), \quad \tilde{\mathbf{y}} = \text{LM}(\tilde{\mathbf{x}}) \quad (4)$$

3. **Update EBM:** Fine-tune using the new pair $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$ and negatives $\mathcal{N}$, with the notation defined previously.

Here, β, β' are learning rates and τ is the temperature. This transfer of information from the energy concept space allows the interpreter to act as a standalone tool for the target LLM.

4 Experiments

We empirically validate our framework by first establishing the pre-trained EBM's efficacy as a faithful surrogate model. We present a representative ablation study to demonstrate its capacity to capture the target LLM's latent semantic relations. Subsequently, we evaluate the interpreter trained on these energy landscapes across two distinct tasks. We emphasize that our framework is designed to train *task-oriented* interpreters; rather than capturing the universal output space of the LLM, each interpreter is specialized for a specific domain. This focused scope allows for effective training even with limited dataset sizes.

4.1 Validating the Dual-Objective EBM

To justify our Dual-Objective framework, we compare three EBM designs selected from our broader experiments. We argue that a robust surrogate must satisfy two requirements: *Fidelity* (adherence to the LLM's output distribution) and *Discriminative Semantic Granularity* (distinguishing causal concepts from noise).

Experimental Setup. Models were trained on the HC3 dataset, a multi-domain Q&A corpus containing both human and model-generated text. We employed `GPT-4o Mini` as the target and `GPT-2 Medium` as the contrastive baseline. We utilized compact 181M parameter models (~71M trainable) on a subset of $20,000$ samples for efficiency. Detailed configurations are in Appendix C. Preliminary scaling experiments demonstrated that larger models achieved wider energy gaps and faster convergence, which correlated with higher accuracy in distinguishing authentic pairs. This indicates the framework's scalability despite the limited scale of this study. We focus our analysis on the following distinct configurations:

- $\mathcal{M}_{\textbf{Fidelity}}$ **(Baseline):** Trained solely on structural integrity ($\lambda = 0$), mimicking standard likelihood modeling.

Table 1: **EBM Ablation Study Results.** We evaluate the models across four dimensions of interpretability. $\mathcal{M}_{\textbf{Fidelity}}$ suffers from the relying on artifacts. $\mathcal{M}_{\textbf{Choice}}$ achieves high margins but creates an abstract latent space that hinders downstream interpretation. $\mathcal{M}_{\textbf{Hybrid}}$ balances robustness with causal precision.

	I. Core Directive (DeYoung et al., 2020)		**II. Robustness** (Gururangan et al., 2018)		**III. SOTA Alignment** (Gemini 2.5 Flash)		**IV. Causal Disentanglement** (Gemini 3 Pro)	
Model	**IR@1** ↑	**SNR** ↑	**Art.** ΔE ↓	**R1E** ↓	**nDCG@3** ↑	**Prec@1** ↑	**Acc** ↑	**ESM** ↑
$\mathcal{M}_{\text{Fidelity}}$	67.62%	1.45	+0.446	85.1%	0.553	26.0%	62.18%	0.145
$\mathcal{M}_{\text{Choice}}$	80.21%	6.89	**-0.014**	**6.9%**	0.699	52.0%	84.16%	**0.499**
$\mathcal{M}_{\textbf{Hybrid}}$	**84.77%**	**7.15**	+0.104	15.17%	**0.796**	**81.0%**	**92.40%**	0.382

- $\mathcal{M}_{\textbf{Choice}}$ **(Ablation):** Trained solely on the Semantic Choice objective ($\lambda = 1$) to identify semantic links across partial segments without overfitting to surface artifacts.

- $\mathcal{M}_{\textbf{Hybrid}}$ **(Ours):** A dual-objective model ($\lambda = 0.9$) using choice samplers for structural dependencies and fidelity signals to regularize the latent space.

Evaluations across four semantic alignments (Table 1) follow protocols in Appendix D.

Dimension I: Core Directive Localization. We assess the ability to localize primary intent amidst background context using Interrogative Recall (IR@1) and Signal-to-Noise Ratio (SNR) (DeYoung et al., 2020). While $\mathcal{M}_{\text{Fidelity}}$ exhibits diffuse attention sensitive to background noise, $\mathcal{M}_{\text{Hybrid}}$ achieves a significant SNR improvement, confirming that the Choice objective effectively prioritizes semantic directives.

Dimension II: Semantic Robustness. We measure sensitivity to spurious "annotation artifacts" (Gururangan et al., 2018) using Artifact Energy Impact and Rank-1 Error (R1E). $\mathcal{M}_{\text{Fidelity}}$ suffers from a fidelity trap, over-prioritizing conversational fillers. Conversely, while $\mathcal{M}_{\text{Choice}}$ successfully ignores artifacts, its abstract latent space led to downstream interpreter collapse in our tests. $\mathcal{M}_{\text{Hybrid}}$ balances this trade-off, retaining the regularization necessary for training.

Dimension III: Alignment with SOTA Oracles. Using `Gemini 2.5 Flash` as a reference oracle, we evaluate ranking quality via nDCG@3 (Järvelin and Kekäläinen, 2002) and Soft Precision@1. $\mathcal{M}_{\text{Hybrid}}$ achieves the highest alignment, indicating that the hybrid objective produces concept rankings most consistent with the generation patterns of state-of-the-art foundation models.

Dimension IV: Causal Disentanglement. We evaluate causal sensitivity via counterfactual triplets (Kaushik et al., 2020) generated by `Gemini 3 Pro`. While $\mathcal{M}_{\text{Choice}}$ produces the sharpest energy landscape, its hyper-discrimination reduces overall accuracy. $\mathcal{M}_{\text{Hybrid}}$ achieves the highest accuracy, successfully balancing discriminative confidence with the robustness required to filter semantic distractors.

4.2 Interpreter Semantic Plausibility Analysis

In the absence of ground-truth labels for concept importance in open-ended generation, we evaluate the *plausibility* of our method by measuring its alignment with the attribution judgments of powerful LLMs. We treat these LLMs as "Oracles," assuming that consensus among diverse high-capacity models serves as a reliable proxy. For this evaluation, we utilize our most robust EBM configuration (see Appendix E for detailed architecture).

Experimental Setup. We interpret the outputs of `GPT-4o Mini` on a subset of 200 prompt-response pairs ($2,000$ combinations) from the HC3-based dataset. For every sentence in the response (the target), we task distinct LLMs (`Gemini 2.5 Flash`, `GPT-4o`, `GPT-4o Mini`, `GPT-J 6B`, and `GPT-2 XL`) to score the sentences in the original prompt based on their contribution to generating that target. We then compare the rankings produced by our Energy-Based Concept-Level Surrogate Interpreter (ESCI) against these Oracles. We report results on two scenarios: *Scenario A* (averaging across all target sentences) and *Scenario B* (focusing on the final sentence, which often contains the conclusion).

Results Analysis. Figure 3 presents the pairwise alignment between our interpreter and the five oracles. We employ *Normalized Discounted Cumulative Gain* (nDCG) to measure ranking quality and *Soft Top-1 Accuracy* (matches if the interpreter's top choice is within the Oracle's top-2) to account

(a) Scenario A: All Targets (Soft Top-1 Accuracy)

Interpreter	Ours	Gemini	GPT-4o	4o-Mini	GPT-J	GPT-2
Ours (ESCI)	**1.00**	**0.75**	**0.75**	**0.64**	**0.79**	**0.70**
Gemini 2.5 Flash	0.67	1.00	0.87	0.75	0.64	0.58
GPT-4o	0.70	0.91	1.00	0.77	0.70	0.68
GPT-4o Mini	0.60	0.85	0.81	1.00	0.66	0.59
GPT-J 6B	0.75	0.69	0.63	0.47	1.00	0.82
GPT-2 XL	0.67	0.71	0.67	0.51	0.87	1.00

(b) Scenario A: All Targets (nDCG Score)

Interpreter	Ours	Gemini	GPT-4o	4o-Mini	GPT-J	GPT-2
Ours (ESCI)	**1.00**	**0.83**	**0.82**	**0.79**	**0.83**	**0.81**
Gemini 2.5 Flash	0.85	1.00	0.89	0.88	0.79	0.75
GPT-4o	0.81	0.91	1.00	0.90	0.84	0.79
GPT-4o Mini	0.77	0.89	0.88	1.00	0.80	0.75
GPT-J 6B	0.85	0.76	0.75	0.74	1.00	0.85
GPT-2 XL	0.81	0.78	0.80	0.77	0.90	1.00

(c) Scenario B: Last Target (Soft Top-1 Accuracy)

Interpreter	Ours	Gemini	GPT-4o	4o-Mini	GPT-J	GPT-2
Ours (ESCI)	**1.00**	**0.83**	**0.82**	**0.71**	**0.97**	**0.92**
Gemini 2.5 Flash	0.77	1.00	0.87	0.75	0.64	0.58
GPT-4o	0.78	0.91	1.00	0.82	0.71	0.59
GPT-4o Mini	0.66	0.85	0.83	1.00	0.68	0.56
GPT-J 6B	0.98	0.69	0.70	0.49	1.00	0.79
GPT-2 XL	0.96	0.71	0.71	0.50	0.85	1.00

(d) Scenario B: Last Target (nDCG Score)

Interpreter	Ours	Gemini	GPT-4o	4o-Mini	GPT-J	GPT-2
Ours (ESCI)	**1.00**	**0.86**	**0.85**	**0.82**	**0.94**	**0.96**
Gemini 2.5 Flash	0.83	1.00	0.93	0.92	0.82	0.78
GPT-4o	0.82	0.95	1.00	0.95	0.86	0.82
GPT-4o Mini	0.81	0.93	0.93	1.00	0.80	0.75
GPT-J 6B	0.92	0.79	0.78	0.77	1.00	0.84
GPT-2 XL	0.90	0.81	0.79	0.77	0.88	1.00

Figure 3: Confusion matrices evaluating interpretation plausibility. Each cell (i, j) represents how well the Interpreter in row i matches the scores of the Oracle in column j. **Soft Top-1** (Left) measures if the Interpreter's top choice appears in the Oracle's top-2. **nDCG** (Right) measures ranking correlation. Our proposed ESCI model (bolded) shows remarkable alignment with GPT-J and GPT-2, suggesting it captures the probabilistic dependencies of standard causal language modeling effectively.

for ambiguity in human-like attribution (see Appendix F for details).

Despite having orders of magnitude fewer parameters ($\sim$71M trainable, $\sim$110M loaded), ESCI achieves competitive plausibility across radically different architectures. In Scenario B (Figure 3c/d), ESCI aligns closely with both `GPT-J 6B`, a standard causal language model, and `Gemini 2.5 Flash`, a heavily instruction-tuned (RLHF) system. This agreement suggests that our energy landscape effectively internalizes the causal mechanics of autoregressive generation, capturing valid semantic links regardless of the specific generation mechanism.

Additionally, we observe that ESCI typically produces sparse, confident attribution scores. In contrast, instruction-tuned Oracles often hedge, distributing probability mass across multiple context sentences. Additionally, the white-box models (`GPT-J`, `GPT-2`) frequently struggled to produce valid probability distributions without significant post-processing intervention. ESCI matches these models in attribution strength while offering a more robust, precise, and user-friendly standalone tool. Our interpreter isolates *necessary* components with a sharper focus on critical dependencies than the often diffuse rankings of generative models.

Qualitative Analysis. Figure 4 provides a granular look at specific attribution behaviors. In cases of clear semantic mapping (Samples 21, 142), ESCI aligns perfectly with Oracles. Disagreements, however, are revealing. In Sample 33, ESCI attributes the simplified output to the topic keyword, whereas Oracles point to the specific user question. This suggests ESCI may sometimes overprioritize the "global topic" over specific query nuances. Conversely, Sample 6 highlights ESCI's strength: the target sentence is a pure stylistic simplification. ESCI correctly identifies the instruction "*Explain like I'm five*" as the cause, whereas Oracles fixate on the semantic content. This confirms ESCI's ability to disentangle stylistic drivers from semantic ones, a critical capability for interpreting instruction-tuned models. Finally, Sample 94 illustrates total chaos, where the target sentence is a high-level analogy synthesizing multiple parts of the prompt, leading to valid but divergent interpretations across all models.

4.3 Evaluating Interpreter Faithfulness

While plausibility confirms that our interpreter aligns with human and oracle judgments, it does not guarantee that the selected concepts are the true drivers of the LLM's generation. To assess this, we evaluate *Causal Faithfulness* by measuring the generative consequences of intervening on the input prompt. We adapt the standard metrics of *Sufficiency* and *Comprehensiveness* (DeYoung et al., 2020), modifying them for open-ended generation by using semantic similarity rather than classification probability (Atanasova et al., 2023).

We define **Generative Sufficiency** as the degree to which the target output can be regenerated using

ID	Prompt Sentences	ESCI	GPT-4o	Gemini	Response Context & Target Sentence
6	[0] Why do people say "half a dozen" instead of "six"? [1] It seems like such a common occurrence. [2] Why take the time and effort to say the extra . . . [3] Explain like I'm five.	0.00 0.00 0.00 **1.00**	**0.70** 0.20 0.10 0.00	**0.80** 0.00 0.00 0.20	Okay! Imagine you have a box of cookies. If you have six cookies, you can just say "six." But if you say "half a dozen," it's like saying "half of a bigger group" of cookies. People like to use "half a dozen" because it sounds a little fancier. . .
21	[0] RGB lines when you take a picture of your monitor . . . [1] Please explain like I'm five.	**0.98** 0.02	**0.80** 0.20	**1.00** 0.00	. . . the camera gets a little mixed up and shows the colors in a funny way. That's why you see those RGB lines! . . .
33	[0] fuel octane. [1] What happens if I feed my Nissan Versa . . . [3] Please explain like I'm five.	**1.00** 0.00 0.00	0.00 **0.70** 0.30	0.00 **0.70** 0.30	. . . not work anymore! So, it's best to stick with the regular gas (87 octane) that your car is designed to use. That way, it will run smoothly and be happy!
48	[0] What those black lines on the road are. [1] EDIT: Sorry about the confusion, I meant . . . [3] Explain like I'm five.	0.00 **0.99** 0.01	0.40 0.10 **0.50**	**0.55** 0.00 0.45	. . . lines show where the lanes are, while others can tell you if you can park or if you need to stop. They are like guides that help everyone follow the rules of the road!
94	[0] What's the point of finding planets light years . . . [2] Why can't we spend money on improving . . . [3] Please explain like I'm five.	**1.00** 0.00 0.00	0.00 **0.70** 0.30	0.30 0.30 **0.40**	. . . while also exploring space, because both are important for our future. It's like making sure your toys are clean and also dreaming about getting new ones!
142	[0] The most prominent members of the current . . . [1] Explain like I'm five.	**1.00** 0.00	**0.80** 0.20	**0.90** 0.10	. . . their own thing and keep the country safe. People are talking a lot about these ideas as they get ready to vote! . . .

Figure 4: Qualitative comparison of attribution scores. **Left:** Prompt snippets. **Middle:** Attribution scores from ESCI and Oracles (Top scores in **bold**, near-zero scores grayed). **Right:** Target sentence from the LLM response. **ID 6**: ESCI correctly attributes the simple tone to the style instruction ("Explain like I'm five"), while Oracles focus on the subject. **ID 33**: ESCI over-focuses on the topic keyword ("fuel octane"), while Oracles correctly identify the specific question. **ID 94**: A case of chaos where the target response is a broad analogy, leading all models to diverge.

only the selected prompt sentences. Conversely, **Generative Comprehensiveness** measures the extent to which the target concept is lost when the selected sentences are *removed* from the prompt. A faithful interpreter should maximize the former and minimize the latter, creating a positive *Faithfulness Gap*. We quantify this using cosine similarity between the embeddings of the original target sentence and the generated counterfactuals.

Table 2 compares our ESCI against Oracles derived from GPT-4o and GPT-4o Mini. The Oracles select sentences based on their own internal importance rankings using a max-ratio thresholding strategy (see Appendix G).

Table 2: **Causal Faithfulness Evaluation.** We measure the semantic similarity of the LLM's response to the target sentence under strict interventions. **Sufficiency:** Prompting with *only* selected sentences. **Comprehensiveness:** Prompting with *everything but* selected sentences. **Gap:** The net causal contribution

Interpreter	Suff. (↑)	Comp. (↓)	Gap (↑)
Ours (ESCI)	**0.409**	**0.214**	**0.195**
GPT-4o (Oracle)	0.409	0.235	0.175
GPT-4o Mini (Oracle)	0.407	0.215	0.192

Analysis. Our proposed ESCI framework achieves *Sufficiency Parity* with the GPT-4o Oracle and slightly outperforms the GPT-4o Mini's self-explanation. This demonstrates that the concept subsets isolated by our method are semantically sufficient to trigger the target response. Critically, ESCI achieves the lowest *Comprehensiveness* score, surpassing both SOTA Oracles. This suggests that our interpreter is more precise in isolating the specific causal antecedents; once our selected sentences are removed, the LLM is significantly less capable of regenerating the target sentence. The resulting *Causal Gap*—the highest among all tested models—confirms that our framework identifies sentences that are not merely relevant, but are the primary necessary drivers of the LLM's output. The residual 0.21 similarity across all models likely represents unavoidable background leakage of domain-specific terminology inherent in the dataset.

5 Conclusion

We introduced a concept-level interpretation framework utilizing Energy-Based Models (EBMs) as surrogates for black-box LLMs. By training a global interpreter on the energy landscape, we achieve sentence-level attribution without requiring inference-time API queries. Our empirical results demonstrate that this approach goes beyond surface-level plausibility; we attained *sufficiency parity* with GPT-4o and superior comprehensiveness, confirming that our model isolates necessary causal drivers while effectively distinguishing true semantic signals from non-causal RLHF priors. Unlike local surrogates, this globally trained approach captures broader generation patterns, offering a scalable pathway for diagnosing model behaviors in high-stakes environments.

6 Limitations and Future Work

Computational Trade-offs. A primary limitation of our framework is the computational overhead of the pre-training phase. Unlike perturbation-based methods (e.g., LIME) which are computationally expensive at *inference* time, our approach shifts this burden to *training*. While this results in a significant one-time cost to train the EBM, it yields a standalone interpreter capable of $O(1)$ inference with zero additional API queries to the target LLM. This makes our method uniquely suitable for high-volume, real-time analysis, though less accessible for users unable to perform the initial pre-training.

Generalizability and Scaling Limits. Our experimental validation focused on standard Q&A tasks using compact surrogate models ($\sim$181M parameters) due to resource constraints. While our scaling experiments suggest improved performance with larger EBMs, the efficacy of the energy landscape in capturing long-range dependencies within massive context windows (e.g., 128k+ tokens) remains to be verified. Future work must investigate the scaling laws of the interpreter network itself to ensure robust attribution in long-context and highly technical regimes.

Benchmarking against White-Box Dynamics. In this study, we prioritized the strict black-box setting common to proprietary APIs, using model-based Oracles (e.g., GPT-4o) as our ground truth. A critical avenue for future work is to apply our framework to open-weights models (e.g., Llama 3, Pythia). Comparing ESCI's energy-based attributions directly against internal white-box signals—such as attention weights or integrated gradients—would provide deeper theoretical insight into how well the surrogate energy landscape approximates the true internal mechanics of the target model.

Human-Centric Evaluation. While our sufficiency and comprehensiveness experiments quantify causal faithfulness, they remain automated proxies. To fully assess the utility of the interpreter, we plan to conduct rigorous human subject studies. These studies will measure the effectiveness of ESCI explanations in aiding users to detect hallucinations, identify bias, and verify the safety of model responses in high-stakes domains.

Downstream Applications and Optimization. Beyond interpretation, we aim to explore the application of ESCI for model improvement. Specifically, the interpreter's ability to identify *necessary* sentences offers a mechanism for **Prompt Optimization**: automatically pruning irrelevant context to reduce token costs without degrading output quality. Additionally, we plan to apply this framework to Chain-of-Thought (CoT) reasoning, using energy scores to verify the causal necessity of intermediate reasoning steps and filter out unfaithful "confabulated" reasoning chains.

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A Preprocessing Details

To ensure compatibility with fixed-dimensional attention mechanisms, we normalize sentence counts during preprocessing. We define a task-dependent hyperparameter, N_{max}, representing the maximum sequence length. After input and output sentences are extracted and embedded, the sequences are padded with a learnable placeholder token or truncated to strictly match this length. This results in dense input tensors $S^{\text{in}}, S^{\text{out}} \in \mathbb{R}^{N_{\text{max}} \times d}$, where d is the embedding dimension of the Sentence-BERT model (768 for `all-mpnet-base-v2`).

B Differentiable Top-K Sentence Selection via Gumbel–Softmax

The interpreter network aims to identify the K most important sentences from the input $\mathbf{x}$ influential in generating the target $\mathbf{y}_T$. Since selecting top-K indices is a discrete, non-differentiable operation, we employ the Gumbel-Softmax relaxation to enable end-to-end training.

Let the interpreter function produce a vector of unnormalized relevance logits $\mathbf{z} \in \mathbb{R}^n$ for the n input sentences, denoted as $z_i = (\mathcal{IN}(\mathbf{x}, \mathbf{y}_T; \alpha))_i$. To introduce stochasticity, we first generate standard Gumbel noise g_i from i.i.d. uniform samples $u_i \sim \text{Uniform}(0, 1)$ as follows:

$$g_i = -\log(-\log u_i), \quad i = 1, \ldots, n. \tag{A1}$$

Given a temperature $\tau > 0$, a single continuous relaxation of a one-hot vector, denoted as $c \in \Delta^{n-1}$, is computed via the softmax function:

$$c_i = \frac{\exp((z_i + g_i)/\tau)}{\sum_{j=1}^{n} \exp((z_j + g_j)/\tau)}, \quad i = 1, \ldots, n. \tag{A2}$$

As $\tau \to 0$, the vector c approaches a discrete one-hot sample from the categorical distribution defined by $\mathbf{z}$. To approximate a K-hot selection vector (selecting multiple sentences), we draw K independent relaxed samples $\{c^{(j)}\}_{j=1}^{K}$ using Eq. A2. We then aggregate these samples by taking their element-wise maximum:

$$m_i = \max_{j=1,\ldots,K} c_i^{(j)}, \quad i = 1, \ldots, n. \tag{A3}$$

The resulting vector $\mathbf{m}$ serves as a continuous proxy for the binary mask. The final output of the interpreter used to gate the input sentences is:

$$\mathcal{IN}(\mathbf{x}, \mathbf{y}_T; \alpha)_i = m_i. \tag{A4}$$

During training, this soft mask allows gradients to backpropagate through the selection process. During inference, we obtain the discrete selection by taking the indices of the top-K logits directly or by hardening the soft mask.

C Energy Network: Pre-training Details

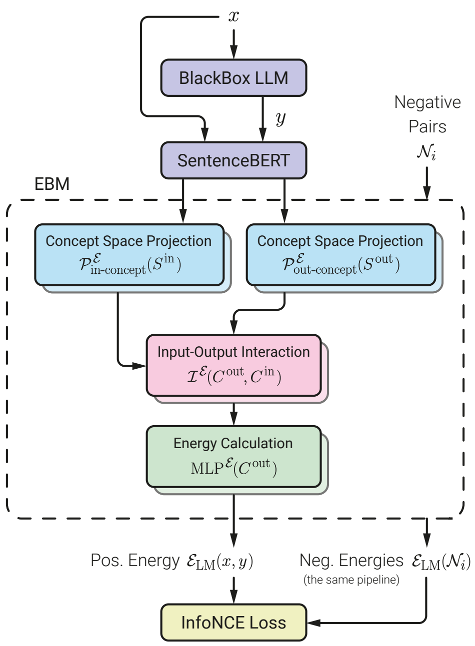

Figure 5: Pre-training pipeline of the globally-aware energy-based surrogate model, $\mathcal{E}_{\text{LM}}(\mathbf{x}, \mathbf{y}; \theta)$. Input and output sentences are first embedded via SentenceBERT and then projected into separate concept spaces through trainable self-attention modules. Input concepts cross-attend to output concepts to model input–output dependencies, after which an MLP computes a scalar energy score. Positive pairs $(\mathbf{x}, \mathbf{y})$ derived from variations of the target LLM's responses and negative pairs $\mathcal{N}_i$ generated via structural and semantic corruptions are both scored, and the model is trained with an InfoNCE loss to assign low energy to authentic pairs and high energy to mismatched ones.

The Energy-Based Model's training pipleine is illustrated in Figure 5. We trained all EBM variants on dual NVIDIA T4 GPUs (provided by Kaggle's free tier) using the AdamW optimizer. The training process required approximately 30 hours. Table 3 details the specific hyperparameters used for the final Hybrid model.

Negative Sampling Strategies. The training objective relies on a diverse set of negative samples to shape the energy landscape. The specific samplers used for each configuration are:

- $\mathcal{M}_{\textbf{Fidelity}}$ **Samplers:**

Table 3: Hyperparameters for the $\mathcal{M}_{\text{Hybrid}}$ EBM.

Parameter	Value
Architecture	
Encoder Model	`all-mpnet-base-v2`
Frozen Parameters	110M
Trainable Parameters	71M
Total Parameters	181M
Projection Dimension (d_{model})	768
Self-Attention Layers	2
Cross-Attention Layers	6
Attention Heads	8
Dropout Rate	0.1
MLP Layers	2
MLP Hidden Factor	2
Optimization	
Epochs	50
Batch Size	16
Learning Rate	$3e^{-5}$
Scheduler	Linear Warmup
Warmup Steps	200
Loss	
Loss Function	InfoNCE
Choice Weight (λ)	0.9
InfoNCE Temperature (τ)	0.1
Margin	0.5
Negative Candidates (K)	5
Data	
Dataset Size	$20,000$ samples
Validation Split	10%
Max Sentence Count	16 (Dynamic Padding)

  - `response_human`: Swaps the LLM response with a human-written answer from the HC3 dataset.
  - `response_other_lm`: Swaps response with a `GPT-2 Medium` output.
  - `response_sentence_masking`: Masks one sentence in the LLM's response randomly.
  - `prompt_sentence_masking`: Masks one sentence in the data pair's prompt randomly.
  - `off_topic`: Swaps response with one from a different prompt in the batch.

- $\mathcal{M}_{\textbf{Hybrid}}$ **Samplers:**
  - `partial_response_choice`: Contrasts a partial prompt against a partial correct response vs. a partial incorrect response.
  - `partial_prompt_choice`: Contrasts a partial response against its true partial prompt vs. a partial off-topic prompt.
  - `response_human`: Same as Fidelity.
  - `response_other_lm`: Same as Fidelity.

D Energy Network: Evaluation Protocols

To rigorously evaluate the EBM's semantic alignment beyond aggregate accuracy, we developed a suite of granular diagnostic tests. This section details the mathematical formulations, dataset filtering criteria, and specific metrics for each testing dimension.

D.1 Dataset Preparation Filtering

For all diagnostic tests, we utilized specific subsets of the HC3 validation set ($N = 1000$). To generate the ground-truth importance scores used for evaluation, we employed an ablation-based energy drop methodology. For each sample pair $(\mathbf{x}, \mathbf{y})$, we systematically removed each sentence to create ablated variants. We calculated the energy for two modes:

- **Prompt Ablation:** Pairs $(\mathbf{x}_{\setminus i}, \mathbf{y})$, where $\mathbf{x}_{\setminus i}$ is the prompt with the i-th sentence removed.
- **Response Ablation:** Pairs $(\mathbf{x}, \mathbf{y}_{\setminus j})$, where $\mathbf{y}_{\setminus j}$ is the response with the j-th sentence removed.

The importance of a sentence was quantified by the positive energy drop caused by its removal relative to the baseline energy $\mathcal{E}(\mathbf{x}, \mathbf{y})$. Using these scored samples, we applied specific filters to isolate relevant linguistic phenomena:

- **Interrogative Subset (**$N = 769$**):** Used for *Dimension I*. We filtered for prompts containing explicit interrogative structures, defined as sentences ending in a question mark or starting with standard interrogative pronouns (e.g., "What", "How", "Why").
- **Artifact Subset (**$N = 890$**):** Used for *Dimension II*. We filtered for responses containing distinct conversational fillers (e.g., "Okay!", "Sure!", "Here is the answer:") appearing as isolated sentences.
- **Oracle Subset (**$N = 500$**):** Used for *Dimension III*. A random subset of validation samples was selected for external scoring by `Gemini 2.5 Flash`.
- **Counterfactual Subset (**$N = 500$**):** Used for *Dimension IV*. `Gemini 3 Pro` was employed to generate specific counterfactual triplets from validation data.

D.2 Dimension I: Core Directive Localization

This test assesses the model's ability to distinguish the primary user intent (the directive) from supplementary context or conversational filler.

Ablation Methodology. For a given prompt $\mathbf{x}$ consisting of n sentences $\{s_1, s_2, \ldots, s_n\}$ and a fixed response $\mathbf{y}$, we calculate the baseline energy $E_{\text{base}} = \mathcal{E}(\mathbf{x}, \mathbf{y})$. We then systematically remove each sentence s_i to create an ablated prompt $\mathbf{x}_{\setminus i}$ and compute the *Relative Energy Impact* (ΔE_i):

$$\Delta E_i = \mathcal{E}(\mathbf{x}_{\setminus i}, \mathbf{y}) - E_{\text{base}} \tag{5}$$

A positive ΔE_i implies that sentence s_i was necessary for the low-energy alignment (i.e., it was semantically "important").

Metric Definitions. Let S_Q be the set of indices corresponding to interrogative sentences and S_{NC} be the set of indices for non-causal context.

- **Interrogative Recall@1 (IR@1):** Adapting the *rationale extraction* evaluation protocol from the **ERASER** benchmark (DeYoung et al., 2020), we define this as the frequency with which the sentence producing the maximum energy impact is an interrogative sentence.

$$\text{IR@1} = \frac{1}{N} \sum_{j=1}^{N} \mathbb{I}\left[{}_{i}(\Delta E_{j,i}) \in S_{Q,j}\right] \tag{6}$$

- **Attribution Signal-to-Noise Ratio (SNR):** Adapting standard signal processing definitions to attribution magnitude, we define this as the ratio of the average energy impact of questions to the average energy impact of non-question context sentences.

$$\text{SNR} = \frac{\frac{1}{|S_Q|} \sum_{i \in S_Q} \Delta E_i}{\frac{1}{|S_{NC}|} \sum_{k \in S_{NC}} \Delta E_k + \epsilon} \tag{7}$$

  where $\epsilon = 1e^{-9}$ is a constant for numerical stability. A high SNR indicates the model is highly sensitive to the directive and insensitive to noise.

D.3 Dimension II: Semantic Robustness

This test measures the model's reliance on non-semantic conversational artifacts, grounded in the concept of "annotation artifacts" in NLI datasets (Gururangan et al., 2018). In generative tasks, models often learn spurious correlations where high-frequency tokens (e.g., conversational fillers) serve as proxies for output validity, regardless of semantic content.

Metric Definitions. Let S_{Art} be the set of indices corresponding to artifact sentences.

- **Artifact Energy Impact (Art. ΔE):** The average change in energy when an artifact sentence is removed. This metric is adapted from Input Reduction methods (Feng et al., 2018), where the goal is to measure the model's sensitivity to the removal of theoretically negligible features.

$$\text{Art. } \Delta E = \frac{1}{|S_{Art}|} \sum_{i \in S_{Art}} \left(\mathcal{E}(\mathbf{x}, \mathbf{y}_{\setminus i}) - E_{\text{base}}\right) \tag{8}$$

- **Rank-1 Error (R1E):** The percentage of samples where an artifact sentence is assigned the highest importance rank (Rank 1). This metric quantifies the fidelity trap, where the model overfits to surface-level plausibility markers rather than semantic drivers.

$$\text{R1E} = \frac{1}{N} \sum_{j=1}^{N} \mathbb{I}\left[{}_{i}(\Delta E_{j,i}) \in S_{Art,j}\right] \times 100 \tag{9}$$

D.4 Dimension III: Oracle Alignment

This test validates the EBM's internal ranking of sentence importance against a "Gold Standard" ranking generated by a SOTA Large Language Model.

Oracle Setup. For each sample in the *Oracle Subset*, `Gemini 2.5 Flash` was provided with the prompt, response, and list of sentences as derived by the EBM, and instructed to assign an integer "Information Density Score" $y_i \in \{1, \ldots, 5\}$ to each response sentence s_i. The scoring criteria were:

- **5 (Critical):** The sentence contains the direct answer or core thesis.
- **1 (Noise):** The sentence is structural fluff or conversational filler.

Metric Definitions. Let $\mathcal{S} = \{s_1, \ldots, s_M\}$ be the set of sentences in a response. Let rel_i be the Oracle's score for sentence i, and let π be the permutation of indices induced by sorting the EBM's energy impact scores ΔE in descending order (i.e., $\pi(1)$ is the index of the most important sentence according to the EBM).

- **nDCG@3 (Normalized Discounted Cumulative Gain):** We measure the ranking quality at cutoff $k = 3$. The Discounted Cumulative Gain (DCG) is computed as:

$$\text{DCG@}k = \sum_{i=1}^{k} \frac{2^{rel_{\pi(i)}} - 1}{\log_2(i+1)} \tag{10}$$

The Ideal DCG (IDCG) is computed similarly using the permutation π^* that sorts the Oracle's scores perfectly. The final metric is:

$$\text{nDCG@}k = \frac{\text{DCG@}k}{\text{IDCG@}k} \tag{11}$$

This metric penalizes the model heavily if it fails to place high-value (Oracle score 5) sentences in the top ranks.

- **Soft Precision@1:** This metric assesses the utility of the single most important sentence identified by the EBM. It is defined as the proportion of samples where the EBM's top choice received a high relevance score (≥ 4) from the Oracle:

$$\text{Soft Prec@1} = \frac{1}{N} \sum_{j=1}^{N} \mathbb{I}\left[rel_{\pi_j(1)} \geq 4\right] \tag{12}$$

D.5 Dimension IV: Causal Disentanglement

This test evaluates the model's ability to identify specific causal links between input and output concepts, distinguishing them from mere topical association.

Counterfactual Setup. For each sample in the *Counterfactual Subset*, the model identified:

1. A specific target response sentence (y_{target}).
2. The high-impact prompt sentence (x_{cause}) that directly necessitated y_{target}.
3. A low-impact distractor sentence ($x_{\text{distractor}}$) from the *same* prompt that was topically related but causally irrelevant to y_{target}.

Metric Definitions.

- **Counterfactual Accuracy:** The percentage of triplets where the EBM correctly assigns lower energy to the causal pair.

$$\text{Acc} = \frac{100}{N} \sum_{j=1}^{N} \mathbb{I}[\mathcal{E}(x_{\text{cause}}, y_{\text{target}}) < \mathcal{E}(x_{\text{distractor}}, y_{\text{target}})] \tag{13}$$

- **Energy Separation Margin (ESM):** The average magnitude of the energy difference between the distractor and the cause. A larger positive margin indicates higher confidence in the causal distinction.

$$\text{ESM} = \frac{1}{N} \sum_{j=1}^{N} (\mathcal{E}(x_{\text{distractor}}, y_{\text{target}}) - \mathcal{E}(x_{\text{cause}}, y_{\text{target}})) \tag{14}$$

E Interpreter: Training Configuration

We report the configuration for the best-performing interpreter, which was trained using the EBM-guided framework on the Hybrid ($\lambda = 0.9$) energy landscape. The training process required approximately 1 hour on dual NVIDIA T4 GPUs (provided by Kaggle's free tier). Table 4 details the hyperparameters.

Table 4: Hyperparameters for the Interpreter Network.

Parameter	Value
Architecture	
Encoder	`all-mpnet-base-v2`
Projection Dim (d_{model})	768
Self-Attention Layers	2
Cross-Attention Layers	6
Attention Heads	8
Dropout Rate	0.1
MLP Layers	1
MLP Hidden Dimension	256
Optimization	
Epochs	50
Batch Size	16
Learning Rate	$1e^{-5}$
Loss Function	InfoNCE ($\tau = 0.1$)
Selection Mechanism	Gumbel-Softmax
Gumbel Temperature	1.0
Data	
Dataset Size	20,000 samples
Validation Split	10%
Max Sentence Count	16 (Dynamic Padding)

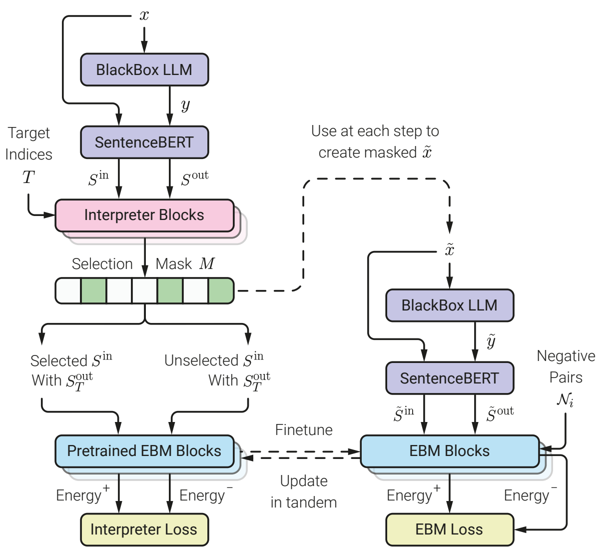

Figure 6: An overview of the EBM-guided training procedure for the interpreter. The framework involves a joint, alternating optimization where the interpreter's parameters are updated using a fixed EBM to create an input mask (left). Subsequently, the EBM is fine-tuned using the LLM's response to the masked prompt to mitigate distribution shift (right).

F Interpreter: Plausibility Evaluation

To construct the plausibility benchmark, we prompted five diverse Large Language Models (`Gemini 2.5 Flash`, `GPT-4o`, `GPT-4o Mini`, `GPT-J 6B`, and `GPT-2 XL`) to act as data annotators.

Prompting Strategy. For a given sample tuple consisting of a prompt $P = \{s_1^p, \dots, s_n^p\}$ and a specific target response sentence s_t^r, each Oracle was provided with the full text context and instructed to: *"Assign an importance score (0.0 to 1.0) to every Prompt Sentence. The scores MUST sum to exactly 1.0."* To maximize determinism, we utilized a temperature of $T = 0$ or close to it.

Metrics. Let $\mathbf{y}_{\text{oracle}} \in \mathbb{R}^n$ be the vector of ground-truth importance scores provided by an Oracle for the n sentences in the prompt. Let $\mathbf{y}_{\text{interp}} \in \mathbb{R}^n$ be the predicted importance scores output by the interpreter.

- **Soft Top-1 Accuracy:** This metric addresses the inherent ambiguity in attribution where multiple prompt sentences may be necessary. We define a match if the interpreter's single highest-scored sentence falls within the top-k sentences identified by the Oracle.

  Let $i^* = \text{argmax}_{i \in \{1,\dots,n\}}(\mathbf{y}_{\text{interp}}^{(i)})$ be the index of the sentence chosen by the interpreter. Let $\mathcal{S}_k(\mathbf{y}_{\text{oracle}})$ be the set of indices corresponding to the k largest values in $\mathbf{y}_{\text{oracle}}$. The metric is defined as:

$$\text{SoftAcc@}k = \mathbb{I}\left[i^* \in \mathcal{S}_k(\mathbf{y}_{\text{oracle}})\right] \quad (15)$$

  In our experiments, we set $k = 2$.

- **nDCG (Normalized Discounted Cumulative Gain):** Similar to EBM experiments, we utilize nDCG to evaluate the quality of the entire ranking order. This metric penalizes the interpreter if it assigns low importance scores to sentences that the Oracle deemed critical.

  Let π be a permutation of indices $\{1, \dots, n\}$ that sorts the interpreter's scores $\mathbf{y}_{\text{interp}}$ in descending order, such that $\mathbf{y}_{\text{interp}}^{(\pi(1))} \geq \mathbf{y}_{\text{interp}}^{(\pi(2))} \geq \dots$. The Discounted Cumulative Gain (DCG) is computed using the Oracle's scores as the true relevance grades:

$$\text{DCG} = \sum_{j=1}^{n} \frac{\mathbf{y}_{\text{oracle}}^{(\pi(j))}}{\log_2(j+1)} \quad (16)$$

  The Ideal DCG (IDCG) is computed similarly using the permutation π^* that sorts $\mathbf{y}_{\text{oracle}}$ in descending order. The final normalized score is:

$$\text{nDCG} = \frac{\text{DCG}}{\text{IDCG}} \quad (17)$$

G Generative Faithfulness Implementation Details

G.1 Metric Definitions

To quantify causal faithfulness in a generative setting, we employ semantic vector similarity. Let $S(\cdot)$ be the embedding function (using `all-mpnet-base-v2`) and $\cos(\mathbf{a}, \mathbf{b})$ be the cosine similarity. For a prompt $\mathbf{x}$, a subset of selected sentences $\mathbf{x}_S$, and a target output sentence $\mathbf{y}_t$:

- **Generative Sufficiency** measures the semantic recovery of the target when prompting with *only* the selected subset:

$$\mathcal{M}_{\text{suff}} = \cos(S(\text{LLM}(\mathbf{x}_S)), S(\mathbf{y}_t)) \quad (18)$$

- **Generative Comprehensiveness** measures the semantic loss when prompting with the complement subset $\mathbf{x} \setminus \mathbf{x}_S$:

$$\mathcal{M}_{\text{comp}} = \cos(S(\text{LLM}(\mathbf{x} \setminus \mathbf{x}_S)), S(\mathbf{y}_t)) \quad (19)$$

G.2 Filtering RLHF Priors (Trivial Targets)

A challenge in interpreting instruction-tuned models is the prevalence of conversational fillers (e.g., "Okay!", "Sure, here is the answer"). These outputs are often driven by Reinforcement Learning from Human Feedback (RLHF) priors rather than specific prompt content. Including them in faithfulness evaluations leads to false negatives for comprehensiveness, as the model will generate them even when the causal source is removed.

We conducted a "Trivial Target Analysis" on a subset of 157 conversational fillers. We found that in **40.8%** of cases, the LLM regenerated the target filler even after the relevant instruction was removed from the prompt. Consequently, we strictly filter these trivial targets from our main evaluation to focus on identifying the causal drivers of semantic content.

G.3 Oracle Baseline Construction

To compare our interpreter against the target LLM's own internal rankings (the Oracle), we employed a Max-Ratio Thresholding strategy. Since the LLM provides continuous importance scores $s_i \in [0, 1]$ for each prompt sentence i, we convert these to the binary selection mask required for sufficiency testing using a dynamic threshold τ:

$$\text{Select } i \iff s_i \geq 0.5 \cdot \max_j(s_j) \qquad (20)$$

This ensures that we capture the primary drivers of the generation while discarding marginal contributors, providing a robust baseline for comparison.

G.4 Robustness Check: NLI vs. Semantic Similarity

We initially evaluated faithfulness using Natural Language Inference (NLI) models (e.g., DeBERTa-v3) to check for logical entailment between the generated counterfactuals and the target. However, we observed that NLI models penalized valid paraphrases too harshly in this generative setting, resulting in near-zero scores for valid causal links (Sufficiency ≈ 0.14). The shift to Cosine Similarity (Sufficiency ≈ 0.41) better captures the "soft" semantic retention characteristic of open-ended generation, while maintaining the same relative ranking between models.

H The LLM Usage

Some parts of the initial drafts of this manuscript were revised with the assistance of a Large Language Model. The model was prompted to improve the fluency, conciseness, and overall academic tone of the text to meet the standards of ACL publications.